

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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(Dated: June 19, 2026)

Quasi-hexagonal (qHP) C₆₀ graphullerene hosts polaronic charge transport in which nearest-neighbor coupling J —not band dispersion—sets mobility scales [1]. Here we quantify how *joint* B/N/P substitution and biaxial strain reshape the *total-energy* landscape of finite qHP supercells, independent of a full Marcus transport closure. PBE+DFT-D3 calculations on tetramers (36 converged neutral points) and periodic $n \times \text{C}_{60}$ cells ($n = 1\text{--}8$; 40 converged points) yield dopant-resolved π -gap evolution, opposite-sign mechanical coefficients α for N (-303 meV/%) vs. B ($+57$ meV/%), and a synergy order parameter $\mathcal{S}$ [Eq. (1)] with $|\mathcal{S}|$ up to ~ 32 meV/atom at $+3\%$ strain. Across $n = 1\text{--}8$ and B/N/P dopants (15 coupled points at $+3\%$), the effects of simultaneous strain and substitution are *not* additive when applied sequentially. At tetramer scale, a 2×2 factorial IPR/ J analysis (pristine/B-doped $\times \epsilon=0/+3\%$) reveals that B substitution dominates delocalization ($\Delta J \approx +20$ meV at $\epsilon=0$), while $+3\%$ tensile strain reduces J under B doping ($46 \rightarrow 37$ meV); all four points remain in the small-polaron regime ($J \ll \lambda/2$).

INTRODUCTION

Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications, yet charge transport in molecular networks is often polaronic [1, 2]. The quasi-hexagonal phase (qHP C₆₀, graphullerene) [2] covalently links C₆₀ units, shortening interfullerene distances by ~ 0.9 Å and raising mobilities by nearly two orders of magnitude over vdW C₆₀ [1]. Capobianco *et al.* [1] showed that the gain tracks enhanced nearest-neighbor coupling J ($\sim 75\text{--}81$ meV vs. 33 meV) within a hopping model, with similar polaron binding ($\lambda^- \sim 0.1$ eV) in both phases.

Beyond pristine monolayers, strain and substitution independently tune qHP C₆₀ band edges [3–5]. Non-additive strain–doping coupling has been reported in other two-dimensional networks [6]; a quantitative energy cross term $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a common PBE+D3 protocol. If formation energies or strain penalties are estimated from *separate* doping-only and strain-only scans, the omitted cross term can reach tens of meV/atom in coupled cells (this work)—large enough to mis-rank dopant stability or strain set-points in screening workflows.

We combine Capobianco-style observables (gaps, IPR, J) with systematic strain–doping energy scans on shared tetramer and size-scaling grids. The present study establishes the *energy* coupling metric $\mathcal{S}$ and band-edge trends; Marcus λ and rate estimates are treated in Supplemental Material and are not required for the non-additivity conclusions here.

COMPUTATIONAL METHODS

Electronic-structure setup

All total-energy and electronic-structure calculations use CP2K [7] with PBE [8] and Grimme DFT-D3 (Becke–Johnson damping) [9], as documented in Supplemental Material. This level is deliberately chosen for a uniform, size-converged scan of (ϵ, δ) ; it differs from the Koopmans-compliant PBE(α_K)+rVV10 framework of Ref. 1, which targets absolute polaron binding and mobilities in pristine supercells.

Rigid-strain approximation. Unless stated otherwise, biaxial strain is imposed by scaling the in-plane lattice vectors of the *pristine* equilibrium cell at *fixed fractional atomic coordinates* (no atomic relaxation under load), not by re-scaling each dopant-relaxed equilibrium lattice. Reported α therefore mixes dopant-induced changes in strain response with any equilibrium lattice offset relative to pristine. This protocol upper-bounds the strain energy and is applied consistently across the comparative grids below. In related 2D carbon networks, allowing ionic relaxation under load typically reduces strain-energy derivatives relative to rigid cells [3]; we therefore interpret reported α and $\mathcal{S}$ as *upper bounds* on the rigid-lattice coupling, with relaxed-ion values expected to be smaller in magnitude but not necessarily vanishing.

Plane-wave cutoff is 300 Ry for cells with $n \leq 5$ fullerene units and 280 Ry for $n \geq 6$ (memory limit for 6×8 cells on our hardware). Because $\mathcal{S}(n)$ compares energies *within* each n at identical cutoff, trends at fixed n are internally consistent. Cross- n comparisons—including the N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ —coincide with the 300→280 Ry switch and should be interpreted cautiously until a matched-cutoff check at

$n = 6$ is available (Supplemental Material Table S2).

SCF convergence criterion 10^{-6} Eh with OT minimization; basis DZVP-MOLOPT-SR-GTH with matching GTH pseudopotentials.

Substitutional doping protocol

Tetramer models ($4 \times C_{60}$, 240 C sites) use substitutional B, N, or P at 5% of carbon sites (~ 12 replacements per cell), chosen by a reproducible pseudo-random protocol (fixed seed 42) over all C coordinates—not a symmetry-averaged or thermodynamic minimum. Periodic size-scaling cells place **one** substitutional dopant per C_{60} unit (fixed count, so nominal concentration $\sim 1/60$ per molecule decreases as n increases). The two protocols differ in *nominal dopant concentration* (5% vs. $\sim 1.7\%$ at $n = 4$); tetramer α , gaps, and IPR/ J are therefore trend indicators, not directly interchangeable with $n=4$ periodic energies without a matched-concentration benchmark. We do not re-optimize dopant sites; all reported α and $\mathcal{S}$ refer to these fixed, reproducible configurations and should be read as site-specific trends pending tests on alternate placements or thermodynamic minima.

Computational grids

In total, $\gtrsim 90$ converged CP2K tasks support the main-text claims (Table S2); the principal neutral-energy grids are:

Mechanical strain response (24 calculations). Pristine and B/N/P tetramers at $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ supply total energies for $\alpha = dE/d\epsilon$ and substitution energies E_f (Table S1).

Electronic structure (12 calculations). The same tetramer compositions at $\epsilon \in \{-5, 0, +5\}\%$ yield π -projected DOS, HOMO–LUMO gaps, and band-edge energies.

Polaron localization (2×2 factorial, four calculations). IPR and nearest-neighbor coupling J are evaluated on a 2×2 grid: pristine vs. B doping crossed with $\epsilon \in \{0, +3\}\%$ (converged neutral single-points; Supplemental Material Fig. S6).

Charged polarons (Supplemental Material). Tetramer GEO.OPT at $q = \pm 1$ and vertical single-points at neutral geometry target Marcus $\lambda^\pm$ [1]; they are not used for $\mathcal{S}$ or α in the main text.

Non-additive size scaling (40 calculations). Periodic cells at $\epsilon \in \{0, +3\}\%$ for each dopant and $n = 1$ –8 supply $\mathcal{S}(n)$; dopant count is one site per C_{60} unit (Table S2).

Reference geometries (Supplemental Material). Neutral tetramer relaxations and a pristine single-point on the relaxed cell define charged-state input geometries.

Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is reported relative to pristine tetramers at the same strain (no chemical-potential reference). In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_f(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the substitution-energy scale: at $n = 4$ and $+3\%$ strain, P has $|\mathcal{S}| \approx 24$ meV/atom while the per-atom substitution scale is $|E_f| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_f| \approx 16$ eV per P site, $n_{\text{dop}}=12$, Table S1)—i.e. $|\mathcal{S}|/|E_f|_{\text{per atom}} \sim 3\%$, large enough to reorder stability margins in sequential doping-then-strain screens even though $|\mathcal{S}|$ exceeds the rigid per-atom strain increment ($\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$ meV/atom for P).

Non-additive coupling metric. We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

To first order in strain, $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$ when $\alpha_\delta = dE/d\epsilon$ is evaluated for the doped cell and α_0 for the pristine reference at the same n (rigid-strain protocol); this links the mechanical coefficient α to the cross term at small ϵ .

Periodic-cell $\mathcal{S}$ values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Supplemental Material Fig. S6). Following Capobianco *et al.* [1], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8), $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$, on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at

fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75$ –81 meV reported for pristine qHP networks [1], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Supplemental Material). Marcus reorganization energies follow the vertical-adiabatic decomposition [1]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5.

RESULTS

Results are organized parallel to Ref. 1: band-edge electronics and localization precede transport-relevant coupling, after which we report mechanical and non-additive energy cross terms specific to coupled (ϵ, δ) tuning.

Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from tetramer single-points at $\epsilon \in \{-5, 0, +5\}\%$. B introduces hole states near the valence manifold (acceptor character), compressing the π gap toward zero at $\epsilon = 0$, whereas N donates electrons into the conduction manifold (donor character) and P (larger covalent radius) perturbs the network more heterogeneously—hence the dopant-specific gap slopes in panel (a). Figure 1(b) shows π -DOS for B @ $\epsilon = 0$.

Polaron localization and electronic coupling

Capobianco *et al.* [1] established that qHP electron polarons remain molecularly localized while J grows with shorter interfullerene distances. A 2×2 factorial separates strain-only, doping-only, and coupled contributions (Supplemental Material Fig. S6): pristine @ $\epsilon=0$ (IPR = 75.0, $J=26.6$ meV); pristine @ +3% (IPR = 76.1, $J=26.5$ meV); B @ $\epsilon=0$ (IPR = 24.6, $J=46.4$ meV); B @ +3% (IPR = 23.1, $J=37.2$ meV). B substitution accounts for most of the IPR drop ($\Delta\text{IPR} \approx -50$) and the zero-strain J enhancement ($\Delta J \approx +20$ meV); +3% strain on pristine is negligible, but under B doping it *lowers* J by ~ 9 meV (negative J cross term). This transport-level non-additivity arises from orbital-coupling sensitivity to interfullerene spacing and is *not* numerically equivalent to total-energy $\mathcal{S}$ (mechanical/electrostatic cross terms in E_{tot}). Using literature $\lambda \sim 0.1$ eV as an order-of-magnitude bound [1], all four points satisfy $J \ll \lambda/2$

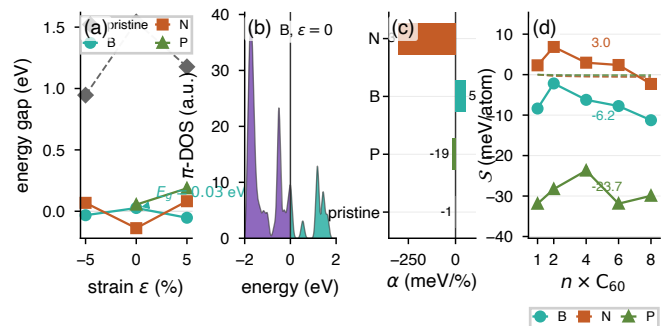


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO–LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon=0$ (Electron SI palette). (c) Mechanical α . (d) $\mathcal{S}(n)$ at +3%; inset: polaron IPR/ J factorial.

(small-polaron hopping); this regime is inferred, not measured here. Figure 1(d) inset and Supplemental Material Fig. S6 summarize the grid; absolute J is below pristine-network values because of fragment size and functional choice.

Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens a wide spread: N ($\alpha = -303.2$ meV/%, tension stabilizes), B (+56.9 meV/%, tension destabilizes), P (−19.1 meV/%) (Table S1). Figure 1(c) highlights the opposite α for N vs. B; ranking dopants by $\epsilon = 0$ stability alone is misleading.

Non-additive strain–doping coupling

Figure 1(d) tracks $\mathcal{S}(n)$ with one dopant per C_{60} . At $n = 4$ and +3% strain, $\mathcal{S} = -6.2$ meV/atom (B), +3.0 meV/atom (N), and −23.7 meV/atom (P); P shows the largest magnitude. Fifteen coupled points span $n \in \{1, 2, 4, 6, 8\}$ and B/N/P at +3%. For N, $\mathcal{S}$ changes sign between $n = 6$ (+2.4 meV/atom) and $n = 8$ (−2.3 meV/atom). Because $n = 6$ –8 use 280 Ry whereas $n \leq 5$ use 300 Ry, this reversal may partly reflect basis-set change rather than intrinsic size physics; we treat it as provisional pending a matched-cutoff $n = 6$ benchmark.

DISCUSSION

Relation to Ref. 1. Capobianco *et al.* [1] compared *pristine* vdW and qHP C_{60} and identified enhanced J —not reduced λ —as the dominant mobility lever, with polaron hopping retained in both phases. Our study transfers that observables vocabulary (gap edges, IPR, J) to *doped* tetramers under strain and adds an en-

ergy cross term $\mathcal{S}$ absent from sequential doping-only or strain-only scans [3, 4]. Where Khan and Leuenberger [4] classify impurity-engineered *half-semiconductor* gaps at fixed geometry, our $\mathcal{S}$ analysis tests total-energy additivity under identical biaxial strain—complementary metrics for impurity–strain co-design [6]. The factorial J pattern (B raises J at $\epsilon=0$ but +3% strain partially reverses it) is qualitatively aligned with Capobianco’s distance–coupling picture: B acceptor states enhance interfullerene hybridization, making J more distance-sensitive, so tensile strain that increases spacing suppresses J far more under B doping than in pristine cells. Transport and total-energy non-additivity are related in origin but not one-to-one; $\mathcal{S}$ should not be used to infer hopping-rate changes without independent J/λ data.

Mechanistic synthesis. (i) **Opposite α for N vs. B.** N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α)—consistent with donor vs. acceptor π -counting and the PDOS shifts in Fig. 1(a,b). (ii) **Large $|\mathcal{S}|$ for P despite small $|\alpha|$.** P’s covalent radius (~ 107 pm vs. ~ 77 pm for C) and longer P–C bonds impose intrinsic cage distortion; rigid external strain couples to this built-in local stress, yielding a cross term that is invisible in separate scans but large when strain and substitutional doping are applied jointly—consistent with P’s intermediate E_f yet largest $|\mathcal{S}|$ at +3%. (iii) **N $\mathcal{S}$ sign change with n .** With one dopant per C_{60} , electrostatic and strain fields are dilute at large n ; the sign flip between $n = 6$ and $n = 8$ may reflect finite-size electrostatics but is confounded with the 280 Ry cutoff introduced at $n \geq 6$ —interpretation is deferred until a constant-cutoff control.

(iv) **Linear $\mathcal{S}$ – α check.** Using tetramer α (Table S1) in $\mathcal{S} \approx (\alpha_\delta - \alpha_0)\epsilon$ at $\epsilon = +3\%$ and dividing by 240 sites gives $\sim +0.7$, -3.8 , and -0.2 meV/atom for B/N/P—far smaller than the periodic $n=4$ values (-6.2 , $+3.0$, -23.7 meV/atom). The gap shows that tetramer-scale linear mechanics and 5% doping do not quantitatively predict periodic $\mathcal{S}$; for P, nonlinear coupling dominates beyond first-order $(\alpha_\delta - \alpha_0)\epsilon$.

Design implications. At +3% strain, P carries the largest additive-approximation error in $\mathcal{S}$ ($|\mathcal{S}| \sim 24$ – 32 meV/atom); co-design should treat P under tension as a *coupled* stability problem, not $|E_f|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$; we do not compute $\mathcal{S}_{\text{gap}}$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone. N $\mathcal{S}$ sign reversal by $n = 8$ cautions against tetramer-only extrapolation.

Limitations and outlook. Rigid strain (upper-

bound on $\mathcal{S}$); fixed-seed substitutional sites without thermodynamic ranking; mixed 280/300 Ry cutoffs across n (N $\mathcal{S}$ sign change at $n = 6$ – 8); tetramer vs. scaling concentration mismatch; PBE+D3 without Koopmans/rVV10; gap non-additivity ($\mathcal{S}_{\text{gap}}$) not quantified; Marcus λ and vertical transitions from charged geometry optimization still pending (Supplemental Material).

CONCLUSIONS

We quantified non-additive strain–doping coupling in heteroatom-substituted qHP C_{60} . Opposite α for N vs. B and finite $\mathcal{S}$ up to ~ 32 meV/atom show that joint (ϵ, δ) tuning is not equivalent to sequential strain and doping scans; for P under tension, the cross term is a few percent of the per-atom substitution-energy scale yet can significantly alter stability rankings in sequential screening workflows. These energy-landscape metrics complement—but do not replace—Marcus λ from charged-state vertical transitions needed to connect $\mathcal{S}$ to polaron hopping rates.

We acknowledge computational resources and support from Xiamen University.

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